

Report on “Interpolating the stochastic heat and wave equations with time-independent noise: solvability and exact asymptotics”

by Le Chen and Nick Eisenberg

Brief description of the main results

In the paper under review, the authors investigate the following (highly) fractional SPDE model:

$$(\partial_t^b + \frac{\nu}{2}(-\Delta)^{a/2})u(t, x) = I^r [\sqrt{\theta} u(t, x) \dot{W}(x)] \quad x \in \mathbb{R}^d, \quad t > 0, \quad (1)$$

where $a \in (0, 2)$, $b \in (0, 2)$, $r \geq 0$, $\nu > 0$, $\theta > 0$, and $\dot{W}$ is a spatial Gaussian noise with mean 0 and covariance given by (the Fourier transform of) a measure of the form $\gamma(x) := \prod_i |x_{(i)}|^{-\alpha_i}$, which covers both the Riesz kernel and the fractional noise situations. The equation is interpreted in the mild Skorohod sense: it thus relies on a convolution with a suitable kernel $G = G_{a,b,\nu,r}$, which is assumed to be nonnegative.

Setting $\alpha := \sum_i \alpha_i$, the main results of the paper can be summed up as follows. If $0 < \alpha < \min(\frac{a}{b}[2(b+r)-1], 2a, d)$, then the equation admits a unique global $L^p(\Omega)$ -solution, for any $p \geq 2$, and in this case, precise asymptotics for the moment $\|u(t, x)\|_{L^p(\Omega)}$ can be derived (relatively to both t and p). On the other hand, if $r \in [0, \frac{1}{2}]$ and $0 < \alpha = \frac{a}{b}[2(b+r)-1] \leq d$, then the equation only admits a (unique) solution up to some specific time T , where its $L^2(\Omega)$ -moment explodes.

By letting the parameters (a, b, r, α_i) vary in (1), the above statements interpolate between several previous results in the literature, whether in the heat case (see e.g. [3, 4, 6]), the wave case (see e.g. [1]) or the fractional-operator case (see e.g. [5]).

The proof strategy leans on a combination of the methods exhibited in the aforementioned literature, together with a close examination of the scaling properties of G .

The central technical property used in the study, and that makes the link with variational constants (namely Theorem 3.5), is an extension of the main result of [2] to the general fractional covariance model described through γ .

Comments and recommendation

The paper is overall well written and nice to read. The many scaling arguments are carried out with great accuracy, and the extension of the technical properties (such as Theorem 3.5) are carefully detailed. I have appreciated the efforts of the authors to illustrate their two main theorems and compare them with previous results in the literature (in Section 2).

Although the global strategy essentially consists in an adaptation of pre-existing methods, I consider that the present results offer a new and interesting perspective on the Malliavin calculus approach to linear multiplicative SPDEs. The interpolation indeed emphasizes how general the procedure, covering both the heat and the wave situations, which is a quite rare phenomenon in SPDE theory.

Thus, in my opinion, this study clearly deserves to be published. Nevertheless, it seems to me that several technical arguments of the analysis require further explanations, and perhaps some modifications.

I have highlighted these various points in an annotated version of the paper.

In brief, I recommend the publication of a revised version of the paper, in which all these technical points will have been carefully treated. Of course, I remain at your disposal for an examination of the revised version.

About the annotated file.

I have used Adobe Acrobat Reader to create this file and make comments through “bubbles”, that appear when clicking the highlights (at least most of them). Please let me know in case you fail to open these bubbles and see the remarks/questions.

References

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- [4] Chen, X., Deya, A., Ouyang, C. and Tindel, S. (2020): Moment estimates for some renormalized parabolic Anderson models. *Ann. Probab.*, to appear 2021.
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